



Digi XBee3 Zigbee 3.0 RF Module

Easy-to-add connectivity in a compact, low-power, low-profile footprint



13 mm x 19 mm

- At 13 mm x 19 mm, the new Digi XBee® 3 micro form factor allows for more compact and portable applications
- Digi XBee 3 is one module for all protocols including: Zigbee, 802.15.4, DigiMesh® and BLE, all configurable via Digi XCTU®
- Eliminate the need for an external microcontroller and create smart end nodes using MicroPython
- Bluetooth® Low Energy for beaconing, connecting to Bluetooth sensors and local configuration using the Digi XBee Mobile app
- Intrinsic IoT security with Digi TrustFence®, a layered approach securing the edge device, through the gateway, into and out of the IoT

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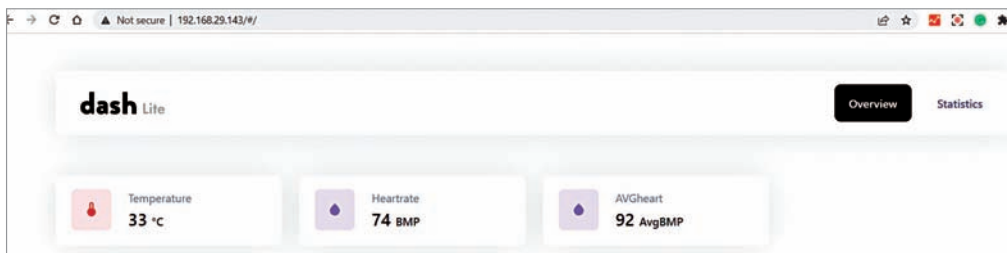


Fig. 7: The device showing real-time health data

If you want to connect your phone to this monitoring device, you can set the Wi-Fi in STA mode. Set the cards' names in the web page UI. You need to show the heart rate, average heart rate, and body temperature data. So, create the cards for these to show in a web page hosted by ESP-DASH.

Now create the setup function and initialise the I2C connection with the Max sensor module and host the UI webpage. In the loop, check the sensor data and update

it in the webpage UI cards. Put the ESP32 in programming mode and connect the ESP chip to FTDI. Select the ESP8266 board in Arduino IDE and select the right port and upload the source code *healthtest.ino*.

Construction and testing

After uploading the source code, solder all the components and connect the battery. Make a small, thin sticker using a sticking tape available at chemists for use on skin to stick it as shown in Fig. 5 and Fig. 6.

Charge the device and slide the switch on the device to on position. You may now stick the IoT health monitor on your forearm or upper arm.

Wait for the device to get connected and then search for the IP address

in the browser. You should be able to see your health data on your phone or any other device you may have connected. Fig. 7 shows how the real-time data looks like.

Bonus. A design with more functions and a market-ready PCB design is available at <https://www.electronicsforu.com/electronics-projects/hardware-diy/smallest-iot-health-sticker> **EFY**

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